

**Research Article****Locality-based community science facilitates early detection and monitoring: the case of an extralimital population of a brachyuran crab**Kevin C.K. Ma^{1,2}, Seshnee Reddy³, P. William Froneman^{1,4}, Francesca Porri^{1,3} and Christopher D. McQuaid¹¹Department of Zoology and Entomology, Rhodes University, Grahamstown, South Africa²Department of Ocean Sciences, Memorial University of Newfoundland, St. John's, Canada³South African Institute for Aquatic Biodiversity, Grahamstown, South Africa⁴Department of Biological Sciences, University of Cape Town, Rondebosch, South AfricaCorresponding author: Kevin C.K. Ma (kevin.ma@mun.ca; kevinckma@gmail.com)

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OPEN ACCESS**Abstract**

Practitioners of community science who are based at a locality and who are familiar with the local ecosystem can provide opportunities to supplement formal monitoring efforts. Between 2021 and 2023, we established a working relationship with a year-round resident recreational fisher who was engaged with monitoring as a community scientist. As a result, the first extralimital record of *Portunus segnis* (commonly known as the blue swimming crab) in temperate waters of South Africa was detected in January of 2020 in the Kowie River on the south coast. Following this discovery, two additional specimens were found in the estuarine reaches of the same river in December 2021. This report provides evidence of a population of *P. segnis* persisting outside its native tropical range and, in addition to morphological identification, species verification was confirmed using genetic barcoding. The circumstances of finding these specimens underscore the importance of establishing and maintaining relationships between local resource users and researchers to facilitate the identification and monitoring of introduced and range expanding species and to provide formal scientific documentation and rapid communication of species occurrences and persistence outside their native range.

Key words: Decapoda, genetic barcoding, Portunidae, *Portunus segnis*, species occurrence, surveillance, volunteer monitoring

Introduction

Invasive species—including extralimital species, defined by Spear and Chown (2008) as species outside their natural geographic range, but within a political jurisdiction—represent a threat to native biodiversity and functioning of ecosystems and to the sustainability of economies at local and global scales (Mack et al. 2000; Mooney and Cleland 2001; Blackburn et al. 2019). Early detection and prevention of new invasions are crucial components of managing risks and impacts (Myers et al. 2000; Rejmánek and Pitcairn 2002; Mehta et al. 2007). Although prevention is preferred, early detection is a vital step that can stop future spread of invasive species because it provides an opportunity for eradication before populations grow larger in size and expand geographically (Simberloff et al. 2005). Early

detection is a cost- and time-effective management option, but it requires high sampling effort to detect incipient invasive populations at the early stages of their establishment (Yemshanov et al. 2019; Ma et al. 2022). Management of invasive species is typically carried out by academic researchers, government agencies, and non-profit organisations (Simpson et al. 2009). They tend to follow established procedures to document the detection of new invasive species, to assess their ecological and socio-economic impacts, and to respond effectively to biological invasions by controlling their spread, mitigating ecological damage, or eradicating them from target areas. Efforts to monitor changes to biodiversity (including new biological invasions) can be supplemented with contributions through community science (Roy et al. 2015; Encarnação et al. 2021). These observations, provided voluntarily, can complement conventional research methodologies and substantially increase the overall temporal and spatial coverage of monitoring. This approach can improve the likelihood of detecting biological invasions, though data from community science can be associated with sampling biases (Probert et al. 2022).

In marine and estuarine systems, crustaceans have frequently been documented as invaders, with brachyuran crabs emerging as among the most aggressive and successful alien species (Brockerhoff and McLay 2011; Swart et al. 2018; Rato et al. 2021; Ben Souissi et al. 2024). In South Africa, four species of introduced brachyuran species have been detected on the west coast: *Carcinus maenas* (Linnaeus, 1758), *Homalaspis plana* (H. Milne Edwards, 1834), *Rathbunixa occidentalis* (Rathbun, 1894) (previously *Pinnixa occidentalis*), and *Xantho hydrophilus* (Herbst, 1790) (Mabin et al. 2017; Griffiths et al. 2018; Peters and Robinson 2018; Robinson et al. 2020). Another non-native species, *Carcinus aestuarii* Nardo, 1847, was also detected on the west coast based on genetic sequencing of collected individuals, but this population has subsequently not been detected since (Geller et al. 1997; Robinson et al. 2005; Mabin et al. 2022). Although not a brachyuran, the anomuran crab, *Porcellana africana* Chace, 1956, is invasive on the west coast (Griffiths et al. 2018; Robinson et al. 2020). Genetic sequencing of megalopa larvae collected from the south coast revealed the presence of *Thalamita gloriensis* Crosnier, 1962, which may be an extralimital brachyuran crab that is native to the Indo-West Pacific, including the east coast of South Africa, and alien to the Mediterranean Sea (Brockerhoff and McLay 2011; Reddy 2022; Govender et al. 2022). Native to the marine and estuarine waters of the east coast of South Africa, the euryhaline brachyuran crab *Portunus segnis* (Forskål, 1775) was recently detected on the south coast as an extralimital species, crossing a major, temperature-defined biogeographic boundary in the Transkei region that separates tropical and temperate waters (Safaie et al. 2013; Emmerson 2016; Ma and McQuaid 2021).

As an unusual observation for estuaries along the south coast of South Africa, a single specimen of *P. segnis* was first caught by a recreational fisher

in January 2020 from the Kowie River. This finding was taxonomically examined in March 2021 and documented in print by November 2021 (Ma and McQuaid 2021). With this occurrence, coupled with a review of all past records of the species in southern Africa—previously reported as *Lupa pelagica* (Linnaeus, 1758), *Lupa pubescens* Dana, 1852, *Neptunus pelagicus* (Linnaeus, 1758), *Portunus pelagica* (Linnaeus, 1758), and *Portunus pelagicus* (Linnaeus, 1758)—a case was made to consider the occurrence of this species in temperate waters to be outside its tropical range (i.e., an extralimital occurrence; Day 1969; Kensley 1981; Lai et al. 2010; Ma and McQuaid 2021). In addition, it was unclear whether this single occurrence represented an individual that was unlikely to survive in this novel environment or an individual from a self-sustaining population at the earlier stages of the range expansion process. To provide evidence of an extralimital population in the Kowie River, we document two additional observations of *P. segnis* made by the same recreational fisher almost two years after its initial detection within the Kowie River, confirming their identity through morphological features and the genetic barcoding of one of these new specimens. Lastly, we discuss the value of locality-based community scientists in providing unusual field observations and physical specimens to researchers, which can ultimately contribute new occurrence records of introduced and range expanding species (e.g., Larson et al. 2020; de Carvalho-Souza et al. 2023).

Materials and methods

Collection of specimens

Two male blue swimming crabs were captured by a recreational fisher with a fishing net in December 2021 some 13.7 km upstream from the mouth of the Kowie River (33°32'46"S, 26°49'15"E). Only one specimen was transported to the laboratory for morphological examination. We examined this specimen (Figure 1) and compared its morphology with other descriptions of *Portunus segnis* (also *Portunus pelagicus* sensu lato) in the literature (Crosnier 1962; Crocetta 2006; Lai et al. 2010; Ma and McQuaid 2021). This specimen was deposited at the Iziko South African Museum, Cape Town, South Africa (catalogue number MB-A094730).

Genetic barcoding

DNA was extracted from a subsample of the muscle tissue from the specimen ($n = 1$) using the salting out protocol (Sunnucks and Hales 1996). A partial fragment of the cytochrome *c* oxidase subunit I (COI) gene was amplified using the primer combination, Crust DF1 and Crust DR1 (Steinke et al. 2016) by polymerase chain reaction (PCR). Following the PCR thermal protocols outlined by Steinke et al. (2016), reactions were set up in a 25 μ l volume containing 12.5 μ l Taq (2x Master Mix, RED, 1.5 mM $MgCl_2$), 6.5 μ l of molecular grade water, 0.5 μ l each of the forward and reverse

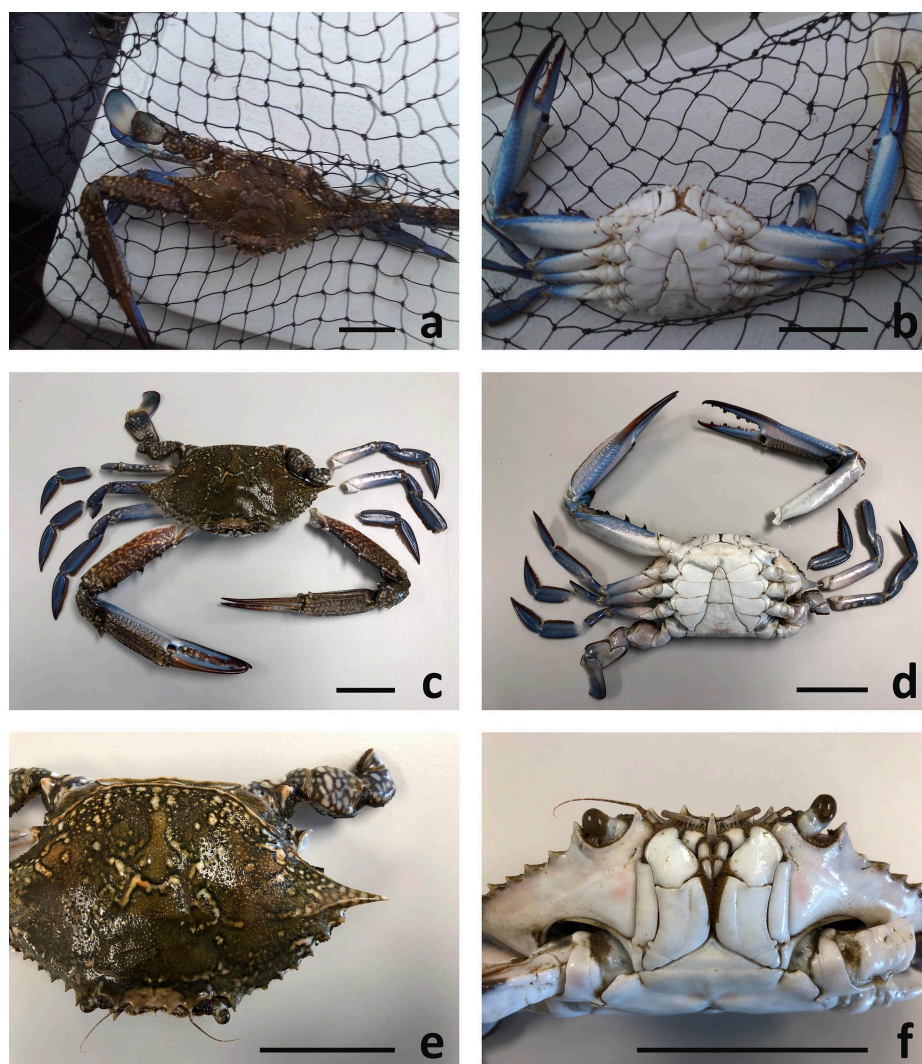


Figure 1. *Ex situ* photographs of an adult male *Portunus segnis* captured from the Kowie River, South Africa: (a) dorsal view at time of collection, (b) ventral view at time of collection, (c) dorsal view, (d) ventral view, (e) dorsal view of the carapace, and (f) frontal view. Scale bar = 50 mm. Credits: (a–b) R. De Cauwer, (c–f) K.C.K. Ma.

primer (10 μ M), and 5 μ l of DNA template (< 100 ng/ μ l, i.e., standardised to create a Master Mix from which the PCR could be run with equal concentrations). Success of the PCR was determined by electrophoresis, followed by visualisation under ultraviolet light. The successful product was then purified using an Exonuclease I – Shrimp Alkaline Phosphatase (Exo/SAP) protocol, and sequenced using BigDye v3.1 (Applied Biosystems) terminator chemistry in the forward and reverse directions and thereafter analysed using an ABI-Hitachi 3500 Genetic Analyser (Applied Biosystems) following the manufacturer's protocol.

The original unedited forward sequence was entered into the Barcode of Life Data (BOLD; <https://www.boldsystems.org/>) identification engine to determine the closest matching species from the reference sequences available on its database (Ratnasingham and Hebert 2007). Species-level identification was assigned to the specimen if the percentage of similarity between the submitted and reference sequence was > 98% (Ko et al. 2013;

Brandão et al. 2016; Azmir et al. 2017). The edited forward and reverse sequences were registered and deposited in GenBank (accession numbers: OR540310 and OR540311).

Results

Species identification

The specimen presented a granulated, greenish-brown carapace of 172 mm in width (measurement included lateral spines) and 83 mm in length, nine teeth (strong lateral spine included) on the arched anterolateral margin, and four triangular teeth (inner supraorbital teeth included) on the frontal margin. The width to length ratio of the carapace was 2:1. Chelipeds were long and slender with a mesiodistal spine on the carpus and three spines on the inner margin of the merus and the right cheliped measured 243 mm in length (manus 136 mm, carpus 37 mm, and merus 114 mm). Although slightly damaged in the examined material, the right natatorial paddle was oval and measured 39 mm in length and 23 mm in width with a length to width ratio of 17:10. The abdomen was triangular and reached the thoracic sternite 4 with the sixth segment of the abdomen wider (18 mm) than the telson (10 mm). Morphological examination of the specimen with respect to its identity was consistent with taxonomic descriptions of *Portunus segnis*, which was previously known as *Portunus pelagicus* sensu lato in southern Africa until the taxonomy of this species complex of four related species was revised by Lai et al. (2010). Further, the extracted DNA produced a successful PCR product and COI sequence. Using BOLD, this sequence generated a 100% match to the family Portunidae and the genus *Portunus* and, at the species level, a 99.67% match to a sequence that was identified as *P. segnis*.

Extralimital population

The observation of adult individuals in 2021 provides limited evidence of an extralimital population of *Portunus segnis* persisting for multiple years in the Kowie River. This 2021 observation was located further upstream from the river mouth compared to the initial observation in 2020; however, it was still downstream from the tidal limit, which is 21 km from the river mouth (Schumann et al. 2001). Although the size of the examined specimen was larger than the earlier 2020 observation, it was also a male (Figure 1). To date, no females have been confirmed from this river.

Discussion

In southern Africa, a number of brachyuran crabs have been undergoing poleward range expansions, spreading from tropical to temperate waters (Branch 1984; Whitfield et al. 2016; Kajee et al. 2018; Peer et al. 2018a, b; Adams et al. 2019; Ma and McQuaid 2021). Contemporary range shifts

may be part of the ongoing global trend of re-distribution of species due to human-mediated climate change (Sorte et al. 2010; Thomas 2010; Sunday et al. 2012, 2015). Monitoring of ecological change in species richness is required to document this re-distribution of species, and can lead to the discovery of species outside their native range (i.e., alien species). In the context of managing biological invasions, preventing introductions of alien species (including modern-day extralimital species) is, of course, preferred. The early detection of alien species, especially nascent populations that are not yet established, is also a key species management goal, allowing rapid mitigation of potential economic and ecological damage in a timely and cost-effective manner (Myers et al. 2000; Rejmánek and Pitcairn 2002; Mehta et al. 2007). Further, extralimital populations should be monitored and managed under the same framework as that of invasive species because their potential to damage economies and ecosystems is easily overlooked. For example, the early detection of extralimital species that are spatially disconnected from their native range (i.e., exhibiting disjunction distributions) can be managed as alien species at early stages of the invasion process and those that are spreading (e.g., contemporary range expansions) can be managed as alien species at later stages, akin to secondary spread (Williamson 2006). At present, extralimital (also known as native-alien) species have been identified as an under-reported ecological threat in South Africa with at least 132 extralimital populations comprising 77 species across marine, freshwater, and terrestrial environments (Nelufule et al. 2023).

A variety of surveying methods can be employed for the purposes of early detection and monitoring in aquatic habitats, including (a) collecting water samples containing microscopic organisms and genetic material (e.g., environmental DNA), (b) installing underwater artificial substrata for sessile and sedentary species to colonise over time, (c) deploying baited and non-baited traps to catch mobile species, (d) utilising SCUBA or remotely operated devices to make *in situ* observations on species, and (e) applying other non-visual approaches such as acoustic telemetry (Ma et al. 2016; Trebitz et al. 2017; LeBlanc et al. 2020; Martinez et al. 2020; Counihan and Thom 2024). In addition to the early detection and monitoring efforts undertaken by researchers and government, contributions of observations from community scientists can result in new records of alien species (Cohen et al. 2011; Azzurro et al. 2013; Bodilis et al. 2013; Moore et al. 2014; Maistrello et al. 2016; Thomas et al. 2017; Jurgens et al. 2018). Through increased sampling effort and spatial coverage, observations from community scientists can substantially increase the probability of detecting new occurrence records that may otherwise be missed. The accuracy of species identification by untrained community scientists remains a major concern, however (Larson et al. 2020; but see Delaney et al. 2008), though some organised community science efforts have been providing appropriate training to address this issue (Gallo and Waitt 2011) and/or implementing other innovations such as machine learning to analyse data (Martinez et al. 2020).

Despite researchers advocating for standardised tools and long-term monitoring (Ma et al. 2020; Loureiro et al. 2021), a long-term, national surveillance strategy for aquatic biological invasions and range expansions in South Africa remains largely elusive. Decades of marine and estuarine monitoring efforts from various sources revealed a current total of 95 alien and 39 cryptogenic species with the crustaceans (e.g., amphipods, barnacles, copepods, decapods, isopods) being the largest represented group, comprising 32% of all recorded alien species (Robinson et al. 2005, 2020; Griffiths et al. 2009, 2010; Mead et al. 2011a, b). Correspondingly, 77 freshwater alien fauna in South Africa were documented in a recent checklist (Weyl et al. 2020).

In the framework of risk management, surveillance for aquatic invasions and range expansions in South Africa could benefit from a Swiss Cheese Model (SCM) to species detection (Reason 1990; Shabani et al. 2024). This approach in South Africa would be especially relevant in the absence of a large-scale, long-term monitoring program, coupled with the dearth/shortage of resident taxonomic experts (Griffiths et al. 2009, 2010; Davies et al. 2020). SCM uses the layering of multiple slices of Swiss cheese as a metaphor for the multiple defenses and barriers to reduce or prevent the risk of accidents within a system (Reason 1990; Shabani et al. 2024). This model can be applied to the early detection and monitoring of aquatic non-indigenous and cryptogenic species in biological systems by using multiple surveillance strategies to minimise the risk of false negatives in species detection (cf. Ekweozor et al. 2016; Kuroshi et al. 2019). At present, the main sources of up-to-date data on the presence and absence (i.e., true positives and true negatives, respectively) of aquatic alien and cryptogenic species in South Africa have been through monitoring and biodiversity studies carried out by researchers in academia, museums, and government agencies (Mead et al. 2011b; Robinson et al. 2020).

Monitoring-related investigations by researchers, including graduate students, suffer from limitations such as sampling at a single timepoint or locality (usually in biodiversity studies) and/or targeting a single alien species (Griffiths et al. 2010; Robinson et al. 2020). Hence, to improve spatial and temporal coverage, monitoring with the assistance of community scientists in South Africa, and elsewhere around the world, appears to be steadily ramping up in recent years (Potts et al. 2021). These efforts can consist of unstructured mass participation in which individuals can directly contribute occurrence records to online databases using smartphone apps and websites (e.g., the *iNaturalist* platform; <https://www.inaturalist.org/>), and structured monitoring in which volunteers follow prescribed sampling procedures (Pocock et al. 2020). The detection of alien and cryptogenic species using this mix of sampling approaches, carried out by researchers and community scientists at irregular intervals across space and time, is imperfect and the risk of false negatives remains high.

Besides mass participation and structured monitoring, independent community scientists can be valuable contributors of initial observations, allowing researchers to conduct follow-up investigations (Potts et al. 2021). This was the case regarding the extralimital records of *Portunus segnis*, which were provided by a locality-based community scientist (incidentally, also a local resource user). Further, this case and the circumstances of its discovery underscore the importance of establishing and maintaining relationships between community scientists and researchers/government (via in-person visits and electronic communication) to facilitate species detection and, eventually, the reporting of introduced and, perhaps even, range expanding species (Ma and McQuaid 2021; Grati et al. 2023). Reporting of some early records of alien species in South Africa may have been similarly aided by individual community scientists (e.g., recreational divers, commercial fishers); examples of contributions of this are known elsewhere in the world (Moore et al. 2014; de Carvalho-Souza et al. 2023). The appeal and importance of community scientists facilitating early detection and monitoring was considered by Larson et al. (2020), who distinguished two main types of community science for early detection: large-scale collection of data resulting in occurrence records to be recorded passively (or ‘mass participation’ as put forward by Pocock et al. 2020) and locality-based research searching for evidence of species introductions or range expansions. Both types were successful in detecting early signs of species introductions and range extensions (see Larson et al. 2020 and references therein).

By appealing to the social aspect of community science through volunteerism and civic responsibility (Andow et al. 2016), its full realisation in South Africa would effectively minimise the risk of false negatives and could shorten the time lag between actual introduction/arrival events and species detections. Put another way, improving/optimising existing SCM mechanisms to early detection and monitoring should improve the likelihood of detecting a novel species occurrence as early as possible, preferably before any ecological or economic damage. Researchers engaged with community scientists in early detection and monitoring efforts may, however, encounter some ethical predicaments, such as destructive sampling methodologies, resulting in the death of native species or damage to the habitat (Pocock et al. 2020). Some possible solutions to address some these ethical issues with regards to early detection may include the recruitment of target interest groups to carry out community science, the formation of technical advisory committees to design and oversee community-based monitoring programs, and the engagement of community scientists to co-design monitoring programs to reduce any conflict of interests (Pocock et al. 2020). Beyond early detection, reporting early signs of a self-sustaining population of a newly detected alien species (including extralimital

species) is also a valuable method of tracking their invasion/range expansion status over time. Ideally, early detection of such self-sustaining populations also allows the re-assessment of any ecological or economic risks they may pose at different time points of their expansion process. The absence of evidence of species persistence following initial detection is the best-case scenario, and this was the case for local disappearance of several alien species in South Africa (Robinson et al. 2005, 2020). In the case of the extralimital spread of *P. segnis* into the Kowie River, the present evidence suggests the persistence of this aggressive predator in the river since its initial discovery, with suspected and known impacts on ecosystems and local fisheries in the Mediterranean Sea where the species has successfully invaded (Castriota et al. 2022; de Carvalho-Souza et al. 2023; Ben Souissi et al. 2024).

Authors' contribution

KCKM: conceptualisation, methodology, formal analysis, investigation, data curation, writing – original draft, writing – review and editing, visualization, project administration; SR: methodology, formal analysis; PWF: resources, writing – review and editing, supervision, funding acquisition; FP: resources, writing – review and editing, supervision; CDM: resources, writing – review and editing, supervision, funding acquisition.

Authors' ORCIDs

Kevin C.K. Ma: [0000-0003-1678-077X](https://orcid.org/0000-0003-1678-077X); Seshnee Reddy: [0000-0002-3221-0753](https://orcid.org/0000-0002-3221-0753); P. William Froneman: [0000-0003-0615-1355](https://orcid.org/0000-0003-0615-1355); Francesca Porri: [0000-0001-8247-4506](https://orcid.org/0000-0001-8247-4506); Christopher D. McQuaid: [0000-0002-3473-8308](https://orcid.org/0000-0002-3473-8308).

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Ethics and permits

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